CB23F35 · LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

B.Tech Computer Science and Business Systems · Rajalakshmi Engineering College

AUDITLENS

QR-Based Machine Safety Inspection and Lockout System

SUBMITTED BY

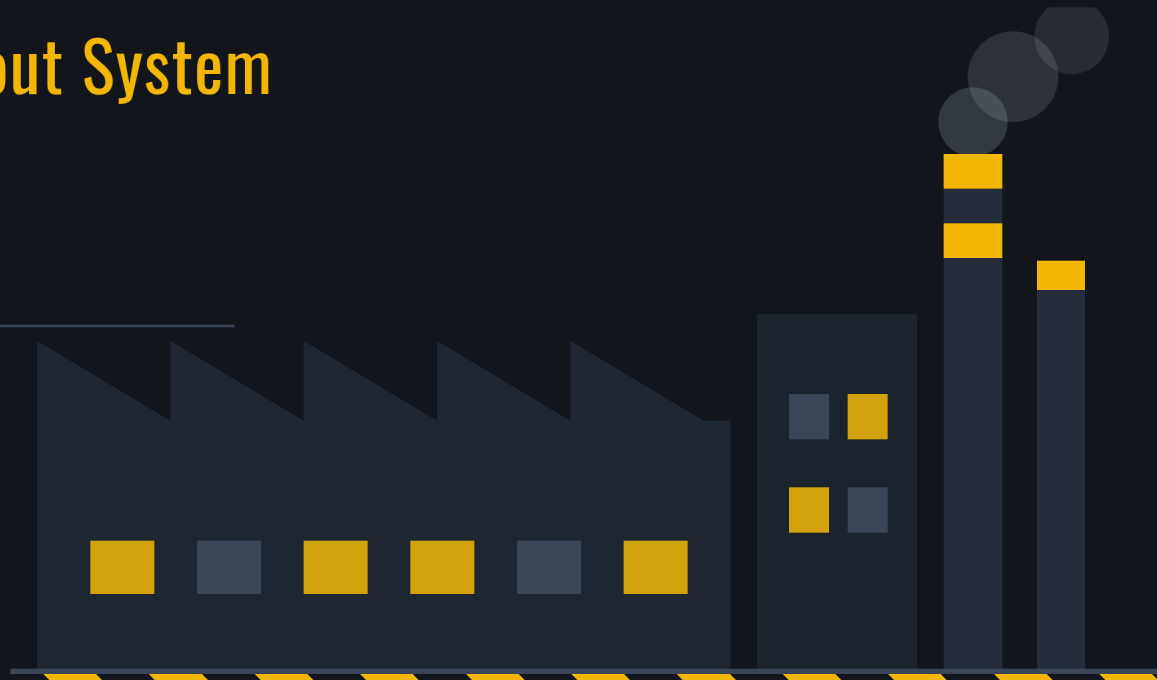
SIVARATHINAM S

Roll No. 2116241401102

FACULTY IN-CHARGE

Mr. M.L. Alphin Ezhil Manuel

Assistant Professor



01 · THE PROJECT

WHAT AUDITLENS DOES

THE PROBLEM

- Paper checklists copied into Excel
- No reminder when a due date passes
- Missing or damaged guards go unnoticed
- Audit proof is slow to find

THE SOLUTION

A low-code app on Zoho Creator. Each machine carries a QR code that opens its live record. Submitting an inspection recalculates the due date and locks out an unsafe machine automatically.

3
Forms

7
Reports

1
Safety dashboard

5
Deluge scripts

02 · WHY IT MATTERS

SAFETY LAW EXPECTS PROOF

INDIA · IN FORCE 21 NOV 2025

OSH Code, 2020

Replaces 13 labour laws, including the Factories Act, 1948. Employers must provide a hazard-free workplace. Inspector-cum-facilitators and third-party audits make records the proof of compliance.

IN AUDITLENS

Dated, searchable inspection records, ready for an inspector or auditor.

INDIA · LEGACY LAW

Factories Act, 1948, Section 21

Dangerous parts of machinery must be securely fenced. The Act is now consolidated into the OSH Code.

IN AUDITLENS

Guard condition (Intact, Damaged, Missing) is checked at every inspection. A missing guard locks the machine out.

USA · 29 CFR 1910.147

OSHA Lockout / Tagout

Hazardous energy must be controlled before anyone services a machine. It is the international reference for lockout practice.

IN AUDITLENS

An unsafe result sets the status to Locked Out and alerts the technician by e-mail.

Sources: Labour Ministry S.O. 5321(E), 21 Nov 2025 · Factories Act, 1948 · OSHA 29 CFR 1910.147

03 · REAL-TIME USE

THREE ROLES, ONE LIVE RECORD

TECHNICIAN

- Scans the QR code on the machine
- Sees status, due date and guard condition
- Submits the inspection in one form

SAFETY MANAGER

- Sees Locked Out, Due Soon and Active counts
- Reads charts by section, risk and hazard
- Gets answers without opening Excel

INSPECTOR

- Filters history by date, machine and result
- Checks certificate number and photo
- Shows that inspections were done and safe

TESTED 13 machines across 5 sections: Packaging, Boiler House, Assembly, Machining and Press Shop. Fits any site with guarded machines and inspection schedules.

04 · FEATURES

SIX THINGS IT DOES FOR YOU

QR CODE PER MACHINE

Scan it to open the live record of that machine.

AUTOMATIC DUE DATES

Monthly 30, Quarterly 90, Half-Yearly 180, Yearly 365 days. Next due date is calculated, never typed.

AUTOMATIC LOCKOUT

An unsafe result or a missing or damaged guard sets Locked Out and e-mails the technician.

DAILY SAFETY CHECK

A scheduled script locks out overdue machines and marks those due within 7 days as Due Soon.

SAFETY DASHBOARD

Four count cards, three bar charts and a guard-condition pie chart.

INSPECTOR REPORT

Inspection history with filters by date, machine and result.

FOUR LAYERS ON ZOHO CREATOR

Layer 1 **USERS**	Administrator · Technician (mobile) · Safety Manager · Inspector
Layer 2 **PRESENTATION**	Forms (3) · Reports (7) · Safety Dashboard · QR link
Layer 3 **LOGIC**	Form workflows (4) · Daily Safety Check · sendmail alerts
Layer 4 **DATA**	Technician · Machine · Inspection_Log

→ **ZOHO MAIL**
Lockout and due-soon e-mails

Layers 2 to 4 run on the hosted Zoho Creator platform.

WHAT WAS BUILT IN ZOHO CREATOR

Component	What it is
Forms	Technician, Machine and Inspection_Log, linked by lookup fields
Reports	4 list, 1 kanban and 2 chart reports, including the Inspector report
Dashboard	Safety Dashboard: 4 count cards, 3 bar charts, 1 guard-condition pie
Deluge	5 scripts: 4 form workflows and the Daily Safety Check schedule
Alerts	Zoho Mail sendmail for lockout and due-soon e-mails
QR code	QR_Code field opens the All Machines report filtered by Serial_Number
Deployment	Development, Stage, then Production: version 1.0, 02 Nov 2026

07 · HOW IT WORKS

FROM SCAN TO LOCKOUT IN FIVE STEPS

1

REGISTER

Admin saves the machine. Days and due date are set, QR code is made.

2

SCAN

Technician scans the QR code. The live machine record opens.

3

INSPECT

Result, guard check, e-stop test, certificate and photo are submitted.

4

DECIDE

Unsafe result or guard not intact: Locked Out. Otherwise: Active.

5

NOTIFY

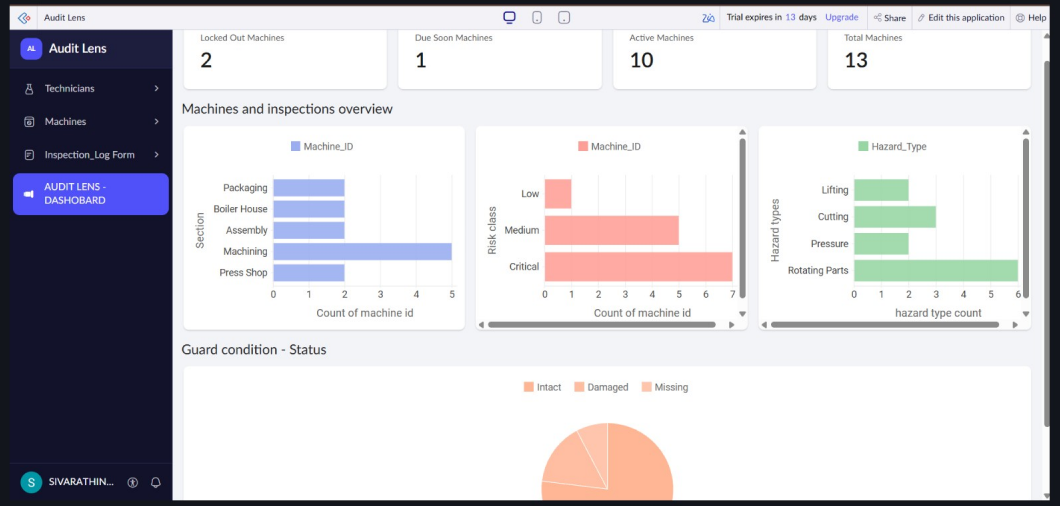
Technician gets an e-mail. Dashboard and reports update.

CORE RULE OF WORKFLOW 4 (SIMPLIFIED)

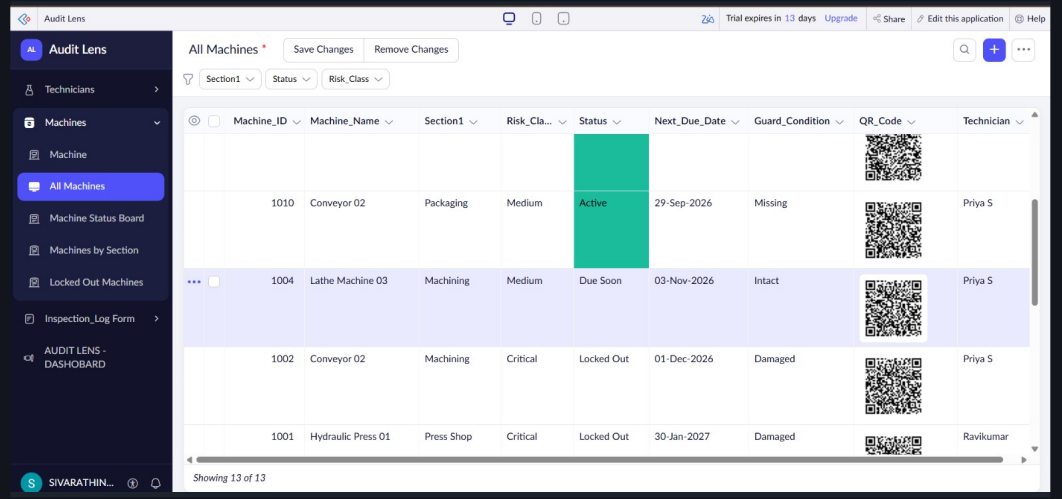
```
Next_Due_Date = Inspection_Date + Frequency_Days;
if (Result == "Unsafe" || Guard_Intact == false)
{ Status = "Locked Out"; sendmail(to: technician); }
else
{ Status = "Active"; }
```

Active	On schedule, guard intact
Due Soon	Due within 7 days
Locked Out	Overdue, unsafe or guard missing

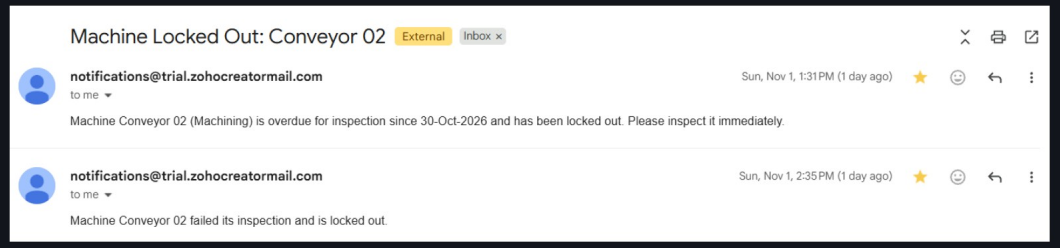
AUDITLENS IN ACTION



Safety Dashboard: live counts and charts



All Machines report: status, due date and QR code



Lockout e-mail received

Conveyor 02 was locked out twice and e-mailed each time: once as overdue, once after a failed inspection.

09 · TESTING

NINE TEST CASES, ALL PASSED

TC	Test	Result	Status
TC-01	Add Quarterly machine	Frequency_Days = 90	Pass
TC-02	Guard Missing on save	Status = Locked Out	Pass
TC-03	Submit Unsafe inspection	Locked Out, dates updated	Pass
TC-04	Check lockout e-mail	E-mail received	Pass
TC-05	Submit Safe inspection	Machine returns to Active	Pass
TC-06	Scan the QR code	Live record opens	Pass
TC-07	Open dashboard cards	13 machines counted	Pass
TC-08	Bulk import 10 machines	5 rejected; pass after fix	Pass*
TC-09	Run Daily Safety Check	Locked Out; Due Soon marked	Pass

9 / 9

test cases passed

8

defects found and closed

Causes: field link names, drop-down spelling, workflow event.

* TC-08: 5 of 10 rows rejected on spelling; all rows imported after the fix · AuditLens · CB23F35

10 · WHAT CAN BE ADDED

FUTURE SCOPE

INSIDE ZOHO CREATOR

Role-based dashboards
Separate views for Technician, Safety Manager and Auditor.

Scheduled summary e-mail
Overdue and due-soon machines sent to the Safety Manager.

“Safe to Operate” page
A public status page for each machine.

WITH INTEGRATIONS

WhatsApp and SMS alerts
Reach technicians where they already read messages.

Zoho Analytics
Deeper trends across sections and hazard types.

LONGER TERM

Predictive maintenance
Use failure history to plan inspections early.

IoT sensor data
Feed guard and machine signals into the record.

Mobile offline mode
Inspect where there is no network.

11 · CONCLUSION

THANK YOU

AuditLens replaces paper and Excel with automatic due dates, lockout and alerts.

Questions are welcome.

SIVARATHINAM S · 2116241401102 · B.Tech CSBS

CB23F35 · Low-Code Application Development using Zoho Creator · Faculty in-charge: Mr. M.L. Alphin Ezhil Manuel

https://creatorapp.zoho.in/sivarathinam.s.2024.csbs1648/audit-lens/